

DATA ANALYSIS PROCESS · STEP 2

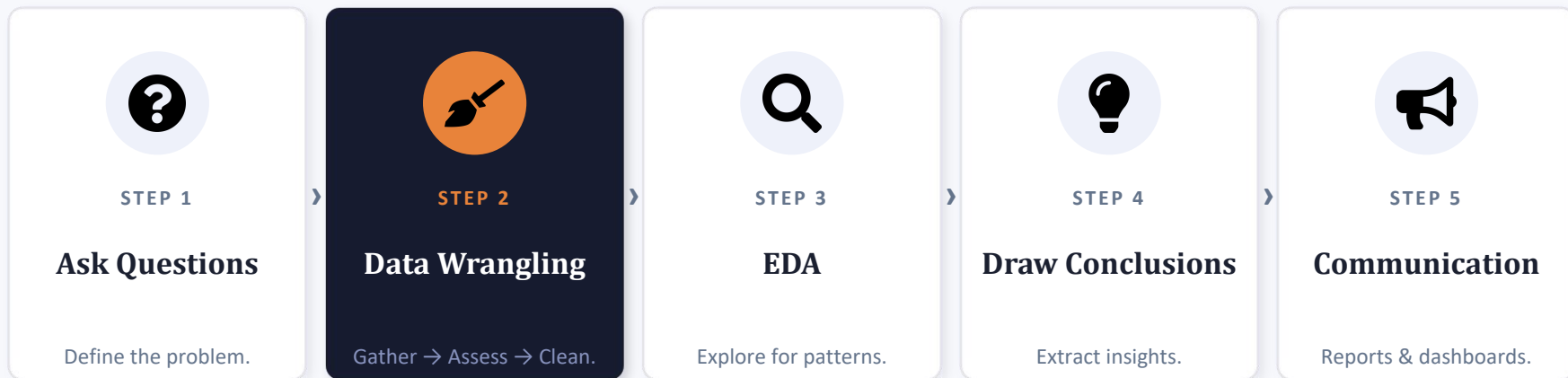
Data Assessing & Cleaning

Finding what's broken in a dataset, and fixing it without breaking something else.

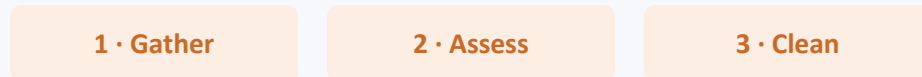
RECAP

The Data Analysis Process

Today's focus: Step 2 — and its three sub-stages.



STEP 2 BREAKS DOWN INTO



Be Sherlock Holmes With Your Data



Real-world data is extremely bad.

Cleaning routinely eats up to 80% of a project's time. Analysts must immerse themselves in the dataset like a detective, hunting for hidden, easy-to-miss flaws before any real analysis can begin.

80%

of a typical data project's time is spent assessing and cleaning — not modeling or visualizing.

Dirty Data vs. Messy Data

The “dirty room” analogy: contents can be bad, or the room itself can be disorganized.



Dirty Data

Quality issues — problems with the content.

- Completeness — is data missing?
- Validity — is it “legal” but wrong?
- Accuracy — is it actually correct?
- Consistency — recorded the same way?



Messy Data

Structural issues — also called Untidy Data.

- 1 Each variable forms a column.
- 2 Each observation forms a row.
- 3 Each observational unit forms a table.

Four Types of Quality Issues

Real examples from the Clinical Trial dataset.

01

Completeness

12 patients are missing address information entirely.

02

Validity

A zip code has only 4 digits instead of 5; a height value is negative.

03

Accuracy

An adult patient's weight is listed as 48 lbs; a name has a typo.

04

Consistency

"New York" vs. "NY"; some names lowercase, others capitalized.

Two Ways to Assess a Dataset

Manual eyes-on-data, paired with programmatic checks.



Manual Assessment

Load the data into Excel or Google Sheets and scroll through it, spotting errors with your own eyes.

Best for spotting things code won't expect — typos, odd formatting, gut-feel anomalies.



Programmatic Assessment

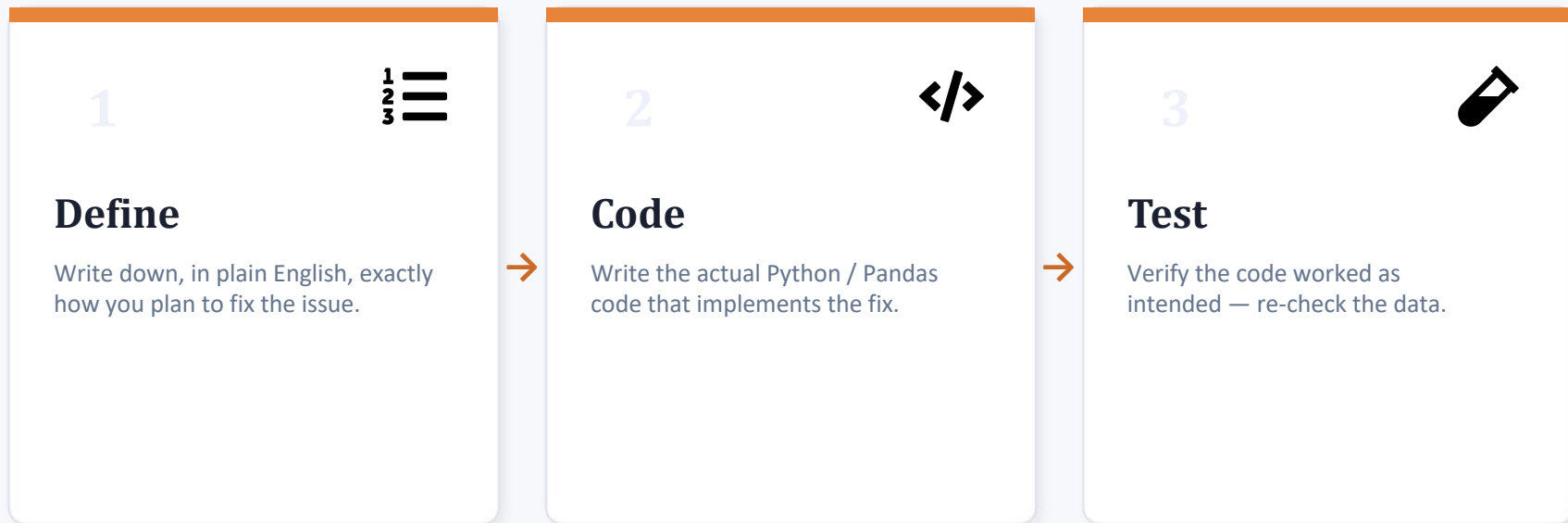
Use Pandas functions to surface issues at scale:

<code>.info()</code>	data types & missing values
<code>.describe()</code>	statistical anomalies, min/max
<code>.sample()</code>	random rows, avoids bias
<code>.duplicated()</code>	repeat entries

FIXING THE PROBLEMS

The Cleaning Workflow — Define, Code, Test

A strict three-step loop, repeated for every single issue found.

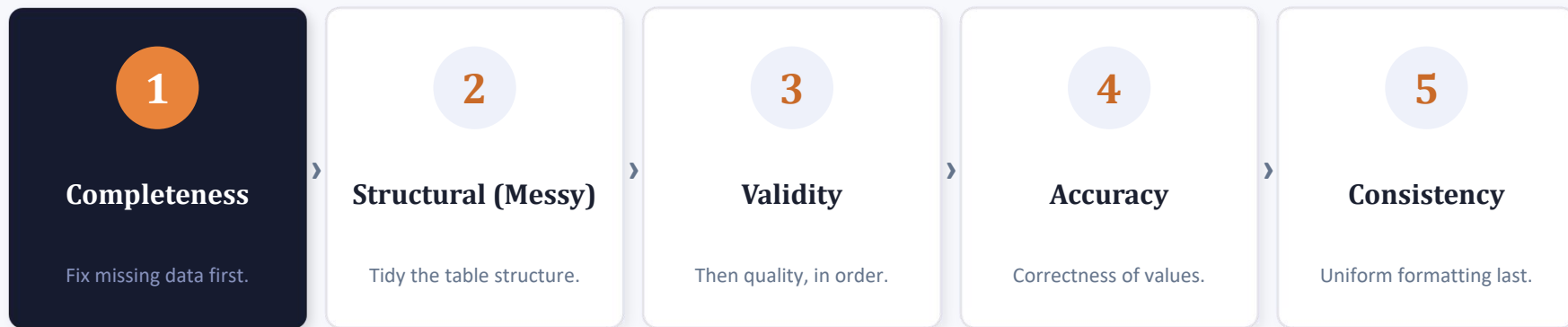


Repeated once per issue — not applied as one big batch fix.

SEQUENCING MATTERS

The Recommended Order of Cleaning

Fix in this order to avoid creating new problems while solving old ones.



Completeness and structure come first because validity, accuracy, and consistency checks depend on a complete, well-shaped table.

WORKED EXAMPLES

Cleaning the Clinical Trial Dataset

Four real fixes applied to the Oralin / Novodra trial data.



Merging Tables

Combined treatments and treatments_cut — same data, split across two files.



Melting Data

`pd.melt()` turned the oralin / novodra columns into rows, satisfying “one variable per column.”



String Manipulation

Split Dosage (“10u - 20u”) into start_dosage and end_dosage, stripping the “u” to get integers.



Correcting Data Types

Converted Assigned Sex to category type — saves memory, improves analysis.

KEY TAKEAWAYS

Clean Data Is the Foundation of Every Insight



Think like a detective

Dirty data hides in plain sight — assess before you trust anything.



Dirty ≠ Messy

Content problems and structural problems need different fixes.



Define → Code → Test

One disciplined loop per issue, never a single batch fix.



Order matters

Completeness and structure first, quality issues last.